Geography Class 6 — Solved Exercises (Complete Book)

Geography (SNC) — English Medium — Class 6 — Punjab Curriculum and Textbook Board (PCTB)

Unit 1 — Structure of Earth and Types of Rocks (book pages 1-19)

Exercise 1 — Circle the correct option (book page 18)

i. The distance in between Earth's surface and its centre is:

12757 km

6371 km ✓ **Correct**

40225 km

40275 km

ii. Average temperature of inner core is:

2000 °C

3000 °C

4000 °C

5000 °C ✓ **Correct**

iii. Ring of fire is located in:

Indian Ocean

Atlantic Ocean

Arctic Ocean

Pacific Ocean ✓ **Correct**

iv. Marble is an example of type of rock:

Igneous

Metamorphic ✓ **Correct**

Sedimentary

Organic

v. Rocks that have changed their properties due to pressure and temperature are called:

Sedimentary Rocks

Metamorphic Rocks ✓ **Correct**

Igneous Rocks

None of these

Exercise 2 — Answer the following short questions (book page 19)

i. Write the names of important layers of the Earth.

The important layers of the Earth are the crust, the mantle and the core. The crust is the thin outer layer, the mantle is the central layer, and the core is the innermost layer, divided into the outer core and the inner core.

ii. Write a short note about global positioning system (GPS).

GPS helps people in positioning and travelling. Signals from different satellites are received on the Earth by control stations, and the user receives that information through a device. It is extremely helpful for tracking companies, travellers, army navigators and tourists.

iii. Enlist the names of the major tectonic plates in the world.

The seven major tectonic plates are the Pacific Plate, the North American Plate, the South American Plate, the African Plate, the Eurasian Plate, the Indo-Australian Plate and the Antarctic Plate. There are also many minor plates besides these seven.

iv. Enlist the names of the commonly found rocks in Pakistan.

Commonly found igneous rocks in Pakistan are obsidian at Chaghi (Balochistan), basalt at Ranikot (Sindh) and gabbro at Zhob (Balochistan).

Limestone, a sedimentary rock, is mostly found in the northern and western mountainous areas of Pakistan, and marble, a metamorphic rock, is also found in the country.

v. Differentiate between extrusive igneous rocks and intrusive igneous rocks.

Intrusive igneous rocks are formed by the solidification of magma in cracks beneath the Earth's surface, for example gabbro, diorite and dolerite.

Extrusive igneous rocks are formed by the solidification of lava at the Earth's surface, where it cools immediately, for example basalt and obsidian.

vi. Write the characteristics of sedimentary rocks.

Sedimentary rocks are formed on the surface of the Earth from weathered and eroded material (sediments) deposited layer by layer. The sediments are transported to far-off places, deposited in layers, and fixed together firmly with the passage of time. Examples are limestone, shale, sandstone, conglomerate and breccia.

Exercise 3 — Answer the following questions in detail (book page 19)

i. How minerals are important for the economy of any country?

Minerals are important for the economy because they are used in industry, construction, medicines, bleaches and soaps, and food preservation. Rocks like gypsum and limestone are used in the cement industry, and rock salt is part of our daily food. A country rich in minerals can develop its industry and trade, and mining itself provides work for many people.

ii. Which types of rocks are located in Pakistan? Discuss their distribution.

All three main types of rocks are found in Pakistan. Igneous rocks include obsidian at Chaghi, basalt at Ranikot and gabbro at Zhob. Sedimentary rocks such as limestone are mostly found in the northern and western mountainous

areas. Metamorphic rocks such as marble and slate are formed where earlier rocks changed under heat and pressure, and the northern mountain areas of Pakistan, where the Eurasian and Indo-Australian plates collide, are rich in such rocks.

iii. What do you know about internal structure of the Earth?

The Earth has three main layers. The crust is the outer layer: continental crust (Sial, silicon and aluminium) is about 40 km deep, and oceanic crust (Sima, silicon and magnesium) is about 8 km deep. The mantle is the central layer, about 2900 km deep; its upper part is molten at about 2000 °C and its lower part is solid at about 3000 °C. The core is the innermost layer: the liquid outer core is about 2250 km deep at about 3800 °C, and the solid inner core is about 1220 km deep at an average temperature of about 5200 °C, made of iron and nickel (Nife).

iv. Discuss modern techniques in the field of geography.

Modern techniques in geography are Geographical Information System (GIS), Remote Sensing (RS), Global Positioning System (GPS) and Google Maps. In remote sensing, satellites collect useful information in the form of images, which are then processed using GIS, making it easier to study inaccessible areas. GPS helps in positioning and travelling through satellite signals, and Google Maps, easily accessible through smart phones, gives information about sites, road networks, traffic conditions and geographical regions.

Critical Thinking Questions (book page 19)

Explore the impacts of modern technology in geography.

Modern technology has made the study of geography faster, more accurate and possible even for inaccessible areas. Satellite images, GIS maps and GPS positioning help in planning, travelling, tracking and emergency services. At

the same time, people must learn to use these tools correctly, because a map or a satellite image is only useful when it is read and interpreted properly.

How rock cycle occur?

The rock cycle is the process of making and breaking of rocks. Weathering and erosion break rocks into small pieces, and agents like rivers, winds and glaciers transport and redeposit the material. The deposited material becomes sedimentary rock, heat and pressure change it into metamorphic rock, and melting and cooling turn it back into igneous rock, so the cycle continues.

Why rocks are important for us?

Rocks are important because they provide building material, cement, salt, chalk and many minerals used in daily life and industry. They also form the soil in which crops grow, and the study of rocks tells us about the history of the Earth. Without rocks there would be no mountains, no soil and no mineral wealth.

How we can compare the Earth with egg?

The Earth can be compared with an egg because both have a thin outer shell, a middle layer and a central part. The eggshell is like the thin crust, the egg white is like the mantle, and the yolk is like the core at the centre. The comparison helps students remember that the crust is very thin compared with the whole Earth, just as the shell is thin compared with the whole egg.

Unit 2 — Mountains, Plateaus and Valleys (book pages 20-32)

Exercise 1 — Circle the correct option (book page 31)

i. Rockies are located in:

Asia

Europe

South America

North America ✓ **Correct**

ii. The highest mountain range in the world is:

Himalayas ✓ **Correct**

Alps

Andes

Rockies

iii. The highest mountain peak in the world is:

Nanga Parbat

Mount Blanc

Mount Everest ✓ **Correct**

K-2

iv. K-2 is also known as:

Godwin Austen ✓ **Correct**

Mount Blanc

Kohsar

Nanga Parbat

v. Rift Valley is formed in which type of mountains:

Residual

Folded

Volcanic

Block ✓ **Correct**

vi. River forms which type of valley:

U-shape

W-shape

V-shape ✓ **Correct**

Y-shape

Exercise 2 — Answer the following short questions (book page 32)

i. Define mountain.

A landmass on the Earth's surface with a rocky, uneven, sloppy and elevated surface is known as a mountain. Mountains rise much higher than the land around them and usually have steep slopes.

ii. What are different types of mountains in the world?

There are three broad types of mountains according to their formation: folded mountains, block mountains and volcanic mountains. Folded mountains are formed by the convergence of tectonic plates, block mountains by the cracking and uplifting of the Earth's crust, and volcanic mountains by layers of lava accumulating and solidifying.

iii. Enlist the names of famous mountain ranges in Pakistan.

Famous mountain ranges in Pakistan are the Himalayas (Lesser and Greater Himalayas), the Karakoram, the Hindukush, the Suleman Range, the Kirthar Range and the Salt Range. The highest peak of the Karakoram is K-2, and the highest peak of the Greater Himalayas in Pakistan is Nanga Parbat (8126 metres).

iv. What is rift valley? Write its example.

A rift valley is a feature associated with block mountains. It is formed when the Earth's crust on both sides of two parallel faults is uplifted or sinks, leaving a low valley between them. Well-known examples are the African Rift Valley and the Great Basin of the USA.

v. Define continental plateau.

A continental plateau is a large, flat, elevated area of land that rises above the surrounding plains. Unlike a piedmont plateau at the foothills of mountains, a continental plateau lies in the interior of a continent and is surrounded by plains or oceans on its sides.

vi. What is meant by V-shape Valley?

A V-shape valley is a valley cut by a river in mountainous areas. The river erodes downwards, so the valley becomes deeper and takes the shape of the English alphabet 'V'. Very deep V-shape valleys are often called gorges, and such valleys are found in the northern areas of Pakistan.

vii. What is the difference between fold and fault?

A fold is a bend in the layers of rock produced when tectonic plates converge and press the crust together; the upper part of a fold is called an anticline and the lower part a syncline. A fault is a crack in the Earth's crust produced by the movement of plates, along which blocks of crust move up, down or sideways.

Exercise 3 — Answer the following questions in detail (book page 32)

i. There are different types of mountains in the world. Discuss each of them.

Folded mountains are formed by the convergence of tectonic plates, which produces folds; the Himalayas, Rockies, Andes and Alps are examples, and the Himalayas hold the world's highest peak, Mount Everest (8848 m). Block mountains are formed when the Earth's crust cracks due to divergence and a block between faults is uplifted; the uplifted block is called a horst, and the sunken part between faults forms a rift valley or graben, for example the Harz Mountains of Germany. Volcanic mountains are formed when lava accumulates and solidifies layer by layer into a conical shape; Vesuvius in Italy and

Fujiyama in Japan are famous examples, and there are three types: cinder cone, composite and shield volcanoes.

ii. What are different types of plateaus? Elaborate.

Plateaus are flat, elevated areas of land, and the book describes piedmont and continental plateaus in particular. Piedmont plateaus are located at the foothills of mountains and are surrounded by mountains on one side and ocean or plain on the other; the Patagonia Plateau in South America and the Potwar Plateau in Pakistan, in the piedmonts of the Himalayas, are examples. The Potwar Plateau lies south of the Siwalik hills, between the rivers Jhelum and Indus, with the Salt Range to its south, and it is rich in oil and gas exploration sites. Continental plateaus are large elevated areas in the interior of continents, rising above the surrounding plains.

iii. What are the impacts of mountains on human life?

Mountains affect human life in many ways. They control climate, because mountain areas are colder and receive more rain and snow, and they are the source of rivers that water the plains. The people of mountain areas work in handicrafts, farming, tourism and hoteling, and beautiful valleys like Naran attract visitors. At the same time, life in mountains is hard because of steep land, cold winters and difficult transport, and mountain areas are also the site of natural hazards like landslides and earthquakes.

iv. Write a descriptive note on the mountains of Pakistan.

Pakistan is surrounded by mighty mountain ranges from the northern and western sides. In the north are the Greater Himalayas, with Gilgit, Swat and Kohistan and the peak Nanga Parbat (8126 m), and north of them the Karakoram range from Gilgit to Skardu, whose highest peak is K-2. The Hindukush, the Suleman Range, the Kirthar Range, the Salt Range and the Toba Kakar mountains lie to the west and south-west. Murree, Galiyat, Kaghan

and Abbottabad are significant valleys in the Himalayas, and glaciers on these mountains continue their creeping movement along the slopes.

Critical Thinking Questions (book page 32)

How physical regions of Pakistan effects the life of the people?

The physical regions of Pakistan shape how people live and work. In the northern and north-western mountainous areas the climate is cold, some areas remain snow-covered all year, and people work in handicrafts, farming, tourism and hoteling. In the plains the climate is extreme, winters cold and summers warm, the soil brought by rivers is favourable for agriculture, and most people depend directly or indirectly on farming.

Explore the impacts of earthquake on ecosystem.

An earthquake shakes and jerks the Earth's crust due to seismic waves, and it damages buildings, roads, water supply and drainage systems. It can cause landslides, change the course of rivers and destroy forests and animal habitats. After an earthquake, the ecosystem takes a long time to recover, because soil, water and vegetation are all disturbed at once.

How Himalayas are important for Pakistan?

The Himalayas are important for Pakistan because they store snow and feed the rivers that water the plains, and they protect the country from cold northern winds. Their valleys, such as Murree, Galiyat, Kaghan and Abbottabad, support tourism, and their forests and minerals support the local economy. They also cause rainfall, because the monsoon winds strike these mountains and bring rain to the country.

Explore importance of valleys for Pakistan.

Valleys are important for Pakistan because they provide flat, fertile land for farming and settlement between mountains, and their rivers provide water for

crops and daily life. Beautiful valleys like Naran attract tourists, which supports hotels and local trade. Valleys also provide routes through mountains for roads and communication between different regions.

Unit 3 — Climatic Regions of the World (book pages 33-45)

Exercise 1 — Circle the correct option (book pages 43-44)

i. Sahara Desert is located in climatic region of:

Equatorial

Tropical ✓ **Correct**

Temperate

Polar

ii. Profession of the people living in tropical area is:

Agriculture ✓ **Correct**

Wood Cutting

Animal Herding

Jobs

iii. Monsoon winds blow in which region:

Equatorial

Tropical ✓ **Correct**

Temperate

Polar

iv. Famous animal of polar climatic region is:

Camel

Reindeer ✓ **Correct**

Lion

Cow

v. People living in Northern Canada are known as:

Pathan

Red Indians

Eskimos ✓ **Correct**

American

vi. Land and sea breezes blow in:

Deserts

Mountains

Coastal Areas ✓ **Correct**

Plateaus

Exercise 2 — Answer the following short questions (book page 44)

i. Define weather and climate.

Weather is the atmospheric condition of temperature, rainfall, air pressure and amount of humidity at a particular place at a particular time. Climate is actually the average of weather conditions over a longer period of time.

ii. Define climatic region.

A climatic region is an area roughly demarcated by lines of latitude, into which the Earth can be divided on the basis of climate. Areas in the same climatic region have similar temperature, rainfall and seasons.

iii. Write the names of different climatic regions of the world.

The different climatic regions of the world are the equatorial region, the tropical region, the temperate region and the polar region. Equatorial areas are hotter, temperate areas are moderate and polar areas are colder.

iv. Define equatorial climatic region.

The equatorial climatic region lies around the equator, where the temperature is high all year and rainfall is heavy. Cocoa and palm are abundant there, but the rubber tree is the most important tree of the equatorial region, and rice, sugarcane, spices, tea, tobacco, banana and coconut are cultivated in its rainforests.

v. What do you know about vegetation in polar climatic region?

Vegetation in the polar climatic region is very poor, because the region is extremely cold and covered with snow and ice for most of the year. Only mosses, lichens and a few small hardy plants can grow in the short summer, and people there live in special houses called igloos.

Exercise 3 — Answer the following questions in detail (book page 44)

i. Explain about weather and climate. Describe the factors affecting the climate of any place.

Weather is the short-term atmospheric condition of a place, while climate is the average of weather over a long period. The climate of a place is affected by its latitude, because equatorial areas are hotter and polar areas colder; by its altitude, because temperature falls with height; by its distance from the sea, because coastal areas remain moderate; and by winds, mountains and rainfall, because mountains like the Himalayas interact with monsoon winds and cause rainfall.

ii. Classify Pakistan into different climatic regions and give brief description of each.

Pakistan is divided into four climatic regions. (1) Sub-Tropical Continental Highland: the northern mountain ranges, the north-western ranges (Chitral, Swat, Dir), the western ranges (Waziristan, Zhob, Loralai) and the mountain

ranges of Balochistan (Quetta, Sarawan, Central Makran, Jhalawan); winters are extremely cold with snowfall, summers quite cool, and rainfall in spring. (2) Sub-Tropical Continental Plateau: most of Balochistan, including Sibbi and Jacobabad; hot and dusty winds blow from May to mid-September, with some rain in January and February. (3) Sub-Tropical Continental Lowlands: the plains of Punjab and Sindh, with hot summers, cold winters and rainfall mainly from the summer monsoon. (4) Tropical Coastland: the coastal area near Karachi, where humidity is high due to the proximity of the sea.

iii. Discuss tropical climatic region in detail.

The tropical climatic region lies between the equatorial and temperate regions, roughly between 30° N and 30° S in the book's forest classification, and it is hot with short, cool winters. Monsoon winds blow in this region, and more than three-quarters of Pakistan's rainfall comes from the summer monsoon. The region is rich in mineral resources, so people are engaged in mining as well as agriculture, and high-quality cotton, wheat, rice and sugarcane are cultivated there. In coastal areas, tropical storms from the Arabian Sea and thunderstorms associated with thermal instability produce some rainfall.

iv. Write the importance of climate in human life.

Climate has significant impacts on human life, because all economic, social, political and commercial activities of people in any area are based on climate. Human activities, lifestyles, customs and traditions vary from place to place with climate, and houses are built differently: roofs are sloppy in areas of heavy snowfall and highly elevated and wide in hotter areas to avoid heat. Climate affects animals too, for example the camel survives in deserts without water for days, and it affects agriculture at large, because temperature, humidity, rainfall and evaporation decide which crops grow where, such as

wheat in the temperate region and rice and rubber in the equatorial and monsoon regions.

Critical Thinking Questions (book page 45)

How climate effects our life?

Climate affects our life because it decides what we wear, what we eat, what kind of house we build and what work we do. Farmers depend on rainfall and temperature for their crops, and traders, travellers and builders all plan their work according to the season. In short, no part of daily life is free from the effect of climate.

Explore the impacts of vegetation on our climate.

Vegetation affects climate because forests give out moisture through transpiration, cool the air and bring rainfall. When forests are cut down, the land becomes hotter and drier, rainfall decreases and soil erosion increases. That is why the book says forests must ideally cover 25% of a country's area for ideal climatic conditions.

Why climatic zones of the world are different?

Climatic zones of the world are different because the Sun's rays do not fall equally on all parts of the Earth. Areas near the equator receive direct rays and are hot, while areas near the poles receive slanting rays and are cold. Latitude, altitude, distance from the sea, winds and mountains add further differences, so each zone has its own temperature and rainfall pattern.

How monsoon effects the climate of Pakistan?

The monsoon greatly affects the climate of Pakistan, because more than three-quarters of the country's rainfall is obtained from the summer monsoon. The monsoon winds bring rain to the plains and to the lower slopes of the Himalayas, which receive maximum rainfall in Pakistan. The northern

mountains, the north-western region and the western part of the plains also get sufficient rainfall from the western disturbance in winter.

Unit 4 — Forests of the World (book pages 46-58)

Exercise 1 — Circle the correct option (book pages 56-57)

i. Forests situated in equatorial and subequatorial areas between 30° N to 30° S are known as:

Tropical Forests ✓ **Correct**

Boreal Forests

Temperate Forests

Taiga

ii. Ideally, forests must cover area of:

50%

40%

25% ✓ **Correct**

10%

iii. There are main types of forests:

3 ✓ **Correct**

5

7

9

iv. Forests above the height of 4000 meters are:

Coniferous

Alpine ✓ **Correct**

Riverine

Rakhs

v. The largest rainforest in the world is:

Indus Forest

Changa Manga

Amazon Forests ✓ **Correct**

Alpine Forests

vi. Area of Pakistan is covered by forests:

5% ✓ **Correct**

10%

25%

50%

vii. Zaire Basin is located in:

Pakistan

Europe

South America

Africa ✓ **Correct**

viii. Chiltan National Park is located in:

Peshawar

Chitral

Gwadar

Quetta ✓ **Correct**

ix. An example of irrigated plantation is:

Alpine

Riverine

Rakhs

Changa Manga ✓ **Correct**

x. The tallest trees are found in:

Emergent Layer ✓ **Correct**

Canopy Layer

Ground Layer

Understory

Exercise 2 — Answer the following short questions (book page 57)

i. Name any two forests found in the world.

Two forests found in the world are the Amazon forests of South America and the Zaire Basin forests of Africa. Both are rainforests in the equatorial belt, and the Amazon is the largest rainforest in the world.

ii. Where temperate forests are located?

Temperate forests are situated in temperate areas, with annual precipitation ranging from 750 mm to 1750 mm. In the book's classification they lie between the tropical and boreal zones, in regions with moderate temperature and four seasons.

iii. How do the forests prevent soil erosion?

Forests prevent soil erosion because the roots of trees hold the soil firmly together, so rain and wind cannot wash it away easily. The leaves and fallen litter also break the force of rain, and the forest floor absorbs water, so less water runs off the surface carrying soil with it.

iv. Name any three forests in Pakistan.

Three forests in Pakistan are the Changa Manga irrigated plantation, the riverine forests along the Indus, and the coniferous forests of the northern mountain areas. The Chiltan National Park near Quetta is also an important protected forest area of the country.

v. Define deforestation.

Deforestation is the elimination of forests to get agricultural or residential land. When trees are cut down on a large scale, the land loses its cover, the climate becomes drier and soil erosion increases.

vi. Evaluate how do the forests boost economic activities?

Forests boost economic activities by providing timber, fuel, medicines, fruits and raw material for paper and furniture industries. They also support tourism and jobs for local people, and by protecting soil and water they support farming, which is the main economic activity of the country.

vii. Write the difference between agriculture and vegetation.

Agriculture is about the growing and harvesting of crops in a particular natural and human environment; plants in agriculture do not grow automatically but are controlled by humans, and agriculture is an activity done by farmers to yield crops. Vegetation is the natural plant cover of an area, such as forests, grasslands, mosses, lichens, algae and shrubs, which grows without human control and is greatly influenced by climate.

viii. What do you know about canopy layer of rainforests?

The canopy layer is formed by tall trees with an average height of 30-40 metres, which are extremely close to each other so that their crowns mix up to form a continuous canopy. This layer does not let sunlight penetrate to the layers below, and non-parasitic plants like lianas and epiphytes are also found in this layer.

Exercise 3 — Answer the following questions in detail (book pages 57-58)

i. Elaborate different layers of rainforests and discuss the features of each layer.

Rainforests have four layers. The emergent layer is the uppermost layer, where the tallest trees rise above the surroundings; their average height is 40-55 metres, they are found in limited numbers, and bats, certain monkeys and butterflies live here. The canopy layer is formed by tall trees of 30-40 metres whose crowns form a continuous cover that blocks sunlight, with lianas and epiphytes growing there. Below it is the understory layer of smaller trees and shrubs growing in shade, and at the bottom is the ground layer of herbs, ferns and fallen leaves on the forest floor.

ii. Name different types of forests in the world. Briefly describe the tropical forests along with characteristics of trees.

There are three main types of forests: tropical forests, temperate forests and boreal forests. Tropical forests are situated in equatorial and sub-equatorial areas between 30° N and 30° S; these areas are hot with short, cool winters, and the season remains almost the same throughout the year, so dense vegetation grows there. The most common type of tropical forest is the evergreen rainforest, found in the equatorial belt including parts of Africa, Southeast Asia and South and Central America, where trees are taller and highly dense, and most rainforest trees are hardwoods.

iii. What are the importance of Amazon rainforest?

The Amazon rainforest is important because it is the largest rainforest in the world and it controls the climate of its region by giving out moisture and absorbing carbon dioxide. It is home to countless species of plants and animals, and its trees provide timber, medicines and oxygen. Its destruction would disturb rainfall, climate and wildlife far beyond South America.

iv. Discuss role of forests in economy of any country in detail.

Forests play a great role in the economy of a country. They provide timber for construction and furniture, fuel wood, paper pulp, medicines, fruits and honey,

and they give work to people in forestry, tourism and related industries.

Forests also protect the soil and water on which farming depends, control floods and climate, and attract tourists, so a country with healthy forests has a stronger and more stable economy.

v. How are the forests distributed all over Pakistan? Write a comprehensive note.

Forests in Pakistan are distributed according to climate and rainfall. Coniferous forests grow in the northern mountain areas, riverine forests grow along the Indus and its tributaries, irrigated plantations like Changa Manga grow in the plains where water is supplied by canals, and scrub forests (rakhs) grow in dry areas. Only about 5% of Pakistan's area is covered by forests, while ideally forests must cover 25% of a country's area, so Pakistan needs to increase its forest cover.

Critical Thinking Questions (book page 58)

How forest are important for our environment?

Forests are important for our environment because they are an essential part of the ecosystem. They clean the air, give out oxygen, bring rainfall, protect soil from erosion and provide shelter for wildlife. Without forests, the climate would become drier, floods would increase and many species would lose their homes.

Explore the impacts of northern (Pakistan) forest on the climate of those areas.

The northern forests of Pakistan make the climate of those areas cooler and wetter, because trees give out moisture and shade the land. They hold snow and rainwater in the soil, feed streams and rivers, and protect the mountain

slopes from erosion. When these forests are cut, the area becomes drier and landslides become more common.

Why forests are different in different areas of Pakistan?

Forests are different in different areas of Pakistan because natural vegetation is greatly influenced by climate, especially precipitation and temperature. Areas with more than 100 cm of rainfall support forests, areas with 25-100 cm support grasslands, and areas with less than 25 cm support only desert vegetation. That is why the north has coniferous forests, the river plains have riverine forests and the dry areas have only scrub.

How we can reduce deforestation in Pakistan?

Deforestation in Pakistan can be reduced by planting new trees for every tree cut down, by protecting existing forests with strict laws, and by giving people alternative fuel and building material so they do not need to cut forests. Public awareness, forest guards, national parks like Chiltan, and irrigated plantations like Changa Manga all help to increase forest cover and protect the environment.

Unit 5 — Natural Disasters (book pages 59-73)

Exercise 1 — Circle the correct option (book pages 71-72)

i. Overflow of water from rivers is called:

Rainfall

Flood ✓ **Correct**

Tsunami

Snowmelt

ii. Result of absence of rainfall for a longer period of time is known as:

Pollution

Earthquake

Heatwave

Drought ✓ **Correct**

iii. World Environment Day is observed on:

5th June ✓ **Correct**

5th December

5th August

5th February

iv. Area of the country must be covered with forests:

10%

15%

25% ✓ **Correct**

50%

v. The most common natural disaster in Pakistan is:

Rainfall

Flood ✓ **Correct**

Tsunami

Snowmelt

vi. A City was demolished due to earthquake in Balochistan (1935):

Quetta ✓ **Correct**

Karachi

Balakot

Islamabad

vii. Landsliding occurs mostly in:

Mountainous Areas ✓ **Correct**

Plain Areas

Coastal Areas

Dry Areas

Exercise 2 — Answer the following short questions (book page 72)

i. Define natural disasters.

Natural disasters are sudden, harmful events caused by natural forces, such as floods, tsunamis, cyclones and earthquakes. They damage life, property, crops and the environment, and they are beyond human control, although human activities can increase their frequency and damage.

ii. Write any three impacts of drought.

Three impacts of drought are: first, crops fail because there is no rain for a long period, so farmers lose their income and food becomes scarce; second, water sources dry up, so people and animals suffer from shortage of drinking water; third, the soil becomes dry and dusty, which increases dust storms and damages the environment.

iii. Write the primary impacts of natural disasters.

The primary impacts of natural disasters are the direct damages they cause at once. Floods bring vast areas of land under water and damage crops, earthquakes shake buildings and break roads and water supply, and landslides block drainage and destroy houses and fields in mountainous areas.

iv. How does global warming cause natural disasters?

Global warming causes natural disasters because the greenhouse effect is increasing in the atmosphere day by day due to human activities. Rising temperature melts glaciers, increases sea level, changes rainfall patterns and makes storms, floods, droughts and heatwaves more frequent and more severe.

v. What is meant by natural disasters?

Natural disasters mean the destructive events of nature that cause great loss of life and property. Examples given in the book are floods, tsunamis, cyclones and earthquakes, and also drought, landslides and volcanic eruptions, which disturb the normal life of an area.

vi. What is meant by three Rs (reduce, recycle, reuse)?

The three Rs mean reduce, recycle and reuse. Reduce means using fewer resources and producing less waste, recycle means processing used material like paper, plastic and glass into new products, and reuse means using the same thing again instead of throwing it away. Following the three Rs saves resources and protects the environment.

Exercise 3 — Answer the following questions in detail (book page 72)

i. What do you know about natural disasters? Write the effects of any three natural disasters in detail.

Natural disasters are sudden natural events that cause great damage to life and property. Floods occur when rivers overflow and bring vast areas under water, damaging crops, houses and roads. Earthquakes are the shaking and jerking of the Earth's crust due to seismic waves; they demolish buildings, block water supply and drainage due to infrastructural damage, and in 1935 an earthquake demolished Quetta in Balochistan. Drought is the absence of rainfall for a long period in an area; it destroys crops, dries up water sources and causes hunger and migration.

ii. Why are natural resources to be preserved? What measures can be taken for preservation of natural resources?

Natural resources must be preserved because they are limited and human life depends on them for food, water, fuel, timber and minerals. Measures for

preservation include planting forests to the ideal 25% cover, using water carefully, following the three Rs (reduce, recycle, reuse), protecting wildlife in national parks, and using renewable energy instead of burning fossil fuels. Awareness and strict laws against overuse also help.

iii. Which types of human activities are responsible natural disasters?

Human activities responsible for increasing natural disasters include cutting down forests, which increases floods, droughts and landslides; burning fossil fuels, which increases the greenhouse effect and global warming; and building on flood plains, river banks and unstable slopes, which increases damage when floods and landslides occur. Pollution of air and water and careless use of land also make the environment weaker against natural hazards.

iv. Global warming is causing increase in frequency and magnitude of natural disasters. How will you justify this statement?

This statement is justified because the greenhouse effect is increasing day by day due to human activities, and the Earth's temperature is rising. Higher temperature melts glaciers and raises sea level, which increases floods and tsunamis; it changes rainfall patterns, causing droughts in some areas and heavy rain in others; and it makes storms and cyclones stronger. That is why floods, the most common disaster in Pakistan, are occurring more frequently now.

Critical Thinking Questions (book page 73)

Why developing countries are the most affected by the natural disasters?

Developing countries are the most affected by natural disasters because they have insufficient early warning systems for natural hazards, as the book notes. Their buildings and infrastructure are weaker, people are poorer and

live in vulnerable areas like flood plains and unstable slopes, and after a disaster they have fewer resources for rescue, relief and rebuilding.

Explore the impacts of rising temperature on our environment.

Rising temperature melts glaciers and snow, raises sea level and changes rainfall patterns, so some areas face floods and others face drought. It increases heatwaves, dust storms and forest fires, damages crops and disturbs wildlife. In Pakistan, rising temperature also increases the melting of mountain snow, which feeds floods in the plains.

Why floods are occurring frequently in Pakistan?

Floods are occurring frequently in Pakistan because of heavy monsoon rains, melting of mountain snow due to rising temperature, and cutting down of forests, which increases run-off. Rivers overflow their banks and bring vast areas under water, and building on flood plains increases the damage. The most common natural disaster in Pakistan is flood, and its frequency is increasing with global warming.

How we can preserve our resources?

We can preserve our resources by using them wisely and without waste: saving water, planting trees, following the three Rs (reduce, recycle, reuse) and using renewable energy. Forests must be protected and increased towards the ideal 25% cover, and laws must stop the overuse of land, water and minerals. Every person can help by reducing waste in daily life.

Unit 6 — Changing Earth and Human Activities (book pages 74-85)

Exercise 1 — Circle the correct option (book pages 83-84)

i. A circular and small part of the sea is called:

Wave

Arches

Bay ✓ **Correct**

Lagoon

ii. Types of weathering are:

2

3 ✓ **Correct**

4

5

iii. Upper part of the wave is called:

Wavelength

Amplitude

Trough

Crest ✓ **Correct**

iv. Rock Eroded from base due to aeolian work is called:

mashroom Rock ✓ **Correct**

Metamorphic Rock

Sedimentary Rock

Organic Rock

v. The most arid province of Pakistan is:

Punjab

Balochistan ✓ **Correct**

Sindh

K.P.

vi. Weathering due to living organisms is known as:

Biological Weathering ✓ **Correct**

Chemical Weathering

Physical Weathering

Exercise 2 — Answer the following short questions (book page 84)

i. Define weathering and erosion.

Weathering is the disintegration of rocks into smaller particles, and the process of weathering is mostly seen in dry and cold areas. Erosion is the carrying away of the weathered material by agents like rivers, glaciers and wind, which continuously do their work and make different types of landforms on the Earth's surface.

ii. Write the names of any four landforms made by coastal erosion.

Four landforms made by coastal erosion are caves, headland and bay, sea arch and stack, and sea cliffs. Continuous wave attack at the bottom of cliffs creates caves, sea water invading inland creates a bay while hard rock remains as a headland, erosion through a headland creates a sea arch which later becomes a stack, and retreating seaward slopes become perpendicular sea cliffs.

iii. What do you know about chemical weathering?

Chemical weathering is the weakening and breaking of rocks by chemical processes. Seawater is saline, and eroded material carried by waves reacts with mineral particles in the water and is dissolved, which weakens the coast; this process is called solution. Acid rain due to air pollution also results in chemical weathering of rocks.

iv. Differentiate arches and stack.

A sea arch is formed when sea waves strike a headland from both sides and continuous erosion creates an opening through the headland. A stack is formed when the sea arch is detached from the headland by further erosion,

so the remaining column-like structure stands alone in the sea; such landforms can be seen in the coastal areas of Pakistan, especially at Hawksbay.

v. Define wind erosion.

Wind erosion is the erosion done by wind, which works notably in arid and semi-arid areas like Balochistan in Pakistan. Wind performs its erosional work in two ways: deflation, carrying unconsolidated sediments with it, and abrasion, using these sediments as a tool for further erosion.

vi. Enlist the agents of weathering.

The agents of weathering and erosion are rivers, glaciers and wind, and also the sea in coastal areas. Heat, rain, wind, rivers and glaciers are the main agents in the formation of sedimentary rocks, and in coastal erosion the four processes involved are corrasion, attrition, solution and hydraulic action.

Exercise 3 — Answer the following questions in detail (book page 84)

i. Discuss different landforms formed by coastal erosion.

Coastal erosion forms several landforms. Continuous wave attack at the bottom of cliffs creates cavities which become caves. Sometimes sea water invades farther inland, creating a bay, while the bordering land of hard rocks remains ahead in the water as a headland. Sea waves striking the headland from both sides create an opening called a sea arch, and when the arch is detached from the headland, the remaining column-like structures are called stacks. On mountainous coasts, seaward slopes exposed by wave attack retreat and become perpendicular sea cliffs.

ii. What are the different types of weathering? Explain each of them.

There are three types of weathering. Physical weathering is the breaking of rocks into smaller pieces by physical forces like heat, cold and pressure, mostly seen in dry and cold areas. Chemical weathering is the breaking of rocks by chemical reactions, such as solution in saline seawater and acid rain due to air pollution. Biological weathering is weathering due to living organisms on the Earth's surface, such as the roots of plants growing into cracks and animals burrowing into rocks.

iii. Elaborate the difference between water erosion and wind erosion.

Water erosion is done by rivers and sea waves. A river breaks rocks in its way, transports the eroded material and deposits it when its speed becomes very low, forming V-shape valleys, waterfalls, pot holes, meanders, ox-bow lakes and deltas. Wind erosion works notably in arid and semi-arid areas like Balochistan: by deflation it carries loose sediments away, and by abrasion it uses those sediments to wear down rocks, forming zeugen, mushroom rocks, deflation basins, yardangs and inselbergs.

iv. Evaluate the role of human activities in erosion of land.

Human activities increase erosion of land in many ways. Cutting down forests removes the roots that hold soil, so rain and wind wash the soil away easily. Ploughing, mining, construction and clearing land for agriculture expose the soil to wind and water, and pollution causes acid rain, which results in chemical weathering. On the other hand, planting trees, building terraces and protecting vegetation can reduce erosion and protect the land.

Critical Thinking Questions (book page 85)

How weathering and erosion are important in formation of soil?

Weathering and erosion are important in the formation of soil because weathering first disintegrates rocks into smaller particles, and erosion then

transports and deposits this material in new places. Over time, the deposited material mixes with organic matter and becomes soil, in which plants can grow. Without weathering there would be no broken rock material, and without erosion there would be no transport of that material to form soil elsewhere.

How we may stop deforestation?

We may stop deforestation by planting new trees for every tree cut, by enforcing strict laws against illegal cutting, and by providing alternative fuel and timber so people do not need to cut forests. Awareness campaigns, national parks and irrigated plantations also help, and every student can help by planting trees in school and at home.

Why our coastal areas are polluting?

Our coastal areas are polluting because waste from cities, factories and ships is thrown into the sea, and oil, plastic and chemicals damage marine life. Air pollution also causes acid rain, which results in chemical weathering of coastal rocks. If waste is not controlled, coastal water, beaches and fishing areas will continue to deteriorate.